

You are required to do Function Point Analysis for the case study given below. This is a real world example of FP Analysis. This will give you hand on experience of performing FP Analysis.

Described below is a description of a small system, a Book shop system.

The following information is maintained by the system

Book details- Book id, Book title, number of copies

Members details- member id, name, address, phone, overdue fines

Rental details- Book id, copy number, member id, return date

The software system needs to provide the following functionality:

to add, change and delete Books from the system

to add, change and delete members' information

to make inquiries on Books using either Book id or Book title

to make inquiries on members

to provide a report of the Books in stock

to provide a report of the members overdue fines

For this problem, do the following:

- a) Identify all ILFs, EIFs, DETs, EIs, EOs, and EQs and calculate their complexities.
- b) Calculate the unadjusted FP count for the application.

Proposed Solution:

ILF's and EIF's

The following information is maintained in the system:

1. Book Details
2. Member Details
3. Rental Details

All the above three are ILFs.

Complexity of ILF's and EIF's

ILF/EIF	RETs	DETs	Complexity	FP Contribution
Book Details	1	< 20	Low	7
Member Details	1	< 20	Low	7
Rental Details	1	< 20	Low	7
Total FP Count				21

Transaction Function Types

Note that maintenance means all four CRUD (Create, Read, Update, Delete) operations are performed on that entity.

Transaction	Transaction Type	FTRs	DETs	Complexity	FP
Add Book	EI	1	<5	Low	3
Update Book	EI	1	<5	Low	3
Delete Book	EI	1	<5	Low	3
Add Member	EI	1	>4,<16	Low	3
Update Member	EI	1	>4,<16	Low	3
Delete Member	EI	1	>4,<16	Low	3
View Book Details	EQ	1	<6	Low	3
View Member Details	EQ	1	<6	Low	3
Calculate Stock	EO	2	<6	Low	4
View Overdue Fines	EQ	1	<6	Low	3
Total FP Count					31

Total Unadjusted function point count for the application = 21+31 = 52